

Muzammil Nawaz Khan

Data Scientist

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SUMMARY

Data scientist with 1+ year of production experience turning messy business data into measurable results. Built a claims-classification model at 95% accuracy across 42,900+ cases, designed a rule-based First-Time Pass Rate analysis by provider, payer, CPT and ICD code that cut the denial rate from 3.2% to 2.4%, and built ARIMA-family models to forecast product prices. Strong in Python, SQL, scikit-learn and gradient boosting, with deep learning and LLM experience. Open to remote work and relocation.

SKILLS

Data & Analysis: Python, SQL, Pandas, NumPy, feature engineering, classification, ARIMA-family forecasting, exploratory data analysis, model evaluation

Machine Learning: Python, PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, XGBoost, AutoGluon, Hugging Face Transformers, Pandas, NumPy

Generative AI & Agents: OpenAI API, Anthropic API, LLMs, prompt engineering, structured outputs, RAG, LLM-as-judge evaluation, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), n8n, ChromaDB, FAISS, custom HNSW, BM25, Reciprocal Rank Fusion

Computer Vision: OpenCV, U-Net / 3D U-Net, SegFormer / SegFormer3D, YOLO, GradCAM, medical image segmentation

Backend & Infrastructure: FastAPI, Django / DRF, REST APIs, Docker, Kubernetes, Git/GitHub, CI/CD

Domain: Healthcare IT, medical imaging, AI training data

EXPERIENCE

Junior AI Engineer, CareCloud (MTBC) Jul 2025 – Aug 2026
Islamabad, Pakistan

- Delivered a production claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% parent-model accuracy and 92%+ category-level accuracy across 42,900+ cases, with daily feedback-loop retraining and multi-month tracking of accuracy and stability.
- Built a Python denial-prediction and resubmission automation system with a rule-based First-Time Pass Rate (FTPR) module comparing rejected vs corrected/paid claims by provider, payer, CPT and ICD code. Cut resubmission wait from 2 weeks to 3 days and reduced the denial rate from 3.2% to 2.4%.
- Layered rule-based checks with an LLM validation agent that generates root-cause analysis and corrective-action recommendations for each flagged case.
- Engineered Python pipelines that pull Hikvision NVR logs, clean them and load them to a database for HR attendance analytics; built an internal FastAPI EDI Viewer for inspecting raw healthcare claim data.
- Designed an n8n workflow acting as a Model Context Protocol (MCP) server for conversation state, backend prompts and call routing, wrapping APIs as tool calls. Cut inference costs 35% through prompt caching.

Software / AI Engineer, Suidhaga Mar 2022 – Nov 2025
Remote, part-time alongside studies

- Applied data analysis and data science to business data; later built ARIMA-family forecasting models to predict product prices.
- Progressed through the full engineering stack over 3+ years for an e-commerce business: built frontend UI, moved to Django-focused backend development, then owned full-stack web features end to end.

AI/ML Research Intern, Risetech Jul 2024 – Aug 2024
Rawalpindi / Islamabad, Pakistan

- Segmented brain tumor regions from multi-modal MRI (T1-CE, T2, FLAIR) on the BraTS20 dataset.
- Benchmarked 3D U-Net against SegFormer3D and selected SegFormer3D (85%+ Dice). Used GradCAM for explainability and FastAPI for serving.
- The work became the Neurofusion final year project: 1st Place at COMPPEC 2025 and a Departmental Silver Medal.

Freelance AI Engineer, Independent Sep 2022 – Present
Remote

- Delivered AI development projects for clients, including model training and applied AI application builds.

Data Trainer, SME Solutions, Inc. (via Deel)

Aug 2026 – Present

Remote, independent contractor

- Evaluate and rank AI model responses to improve accuracy in different contexts.
- Test models for inaccuracies and bias, validating their applicability in the target domains.

PROJECTS

Freight Rates Predictor (Python, scikit-learn): End-to-end ML pipeline on spatial, load, market-index and temporal features.

GitHub: github.com/muzammil5539/Machine-Learning-Engineer-Assessment-Ena-Spotter

Fuel Route Optimizer (Django, OSRM): Geocoded about 8,150 fuel stations, OSRM routing, an 8-mile corridor filter and a Dijkstra cheapest-fuel plan with a 500-mile range; regression-tested. GitHub: github.com/muzammil5539/django_application

Document Summarizer (FastAPI, Next.js 14, LangChain, ChromaDB): Automatic strategy router (Stuff, Map-Reduce, Refine, Recursive), SSE token streaming and async pipelines. About 50% lower latency and 99% factual consistency.

RAG Custom Engine (Pure Python, GPT-4o-mini): From-scratch hybrid retrieval with a custom HNSW store, Okapi BM25 and Reciprocal Rank Fusion, plus a 3-stage Self-RAG gate (retrieval decision, relevance grading, hallucination check) and contextual compression. About 20% precision gain over dense-vector baselines.

EDUCATION

Bachelor of Computer Engineering, National University of Sciences and Technology (NUST)

Nov 2021 – Jul 2025

Islamabad, Pakistan

CGPA 3.25. Final year project: Neurofusion, 3D brain tumor segmentation (SegFormer3D, BraTS20, 85%+ Dice); 1st Place COMPPEC 2025, Departmental Silver Medal.

F.Sc. Pre-Engineering, Ghazali Premier College

Jun 2018 – Aug 2020

Lahore, Pakistan

CERTIFICATIONS

- IBM AI Engineering Professional Certificate
- IBM Deep Learning with PyTorch, Keras and TensorFlow Professional Certificate
- Complete Modern C++ (C++11/14/17), Packt
- Generative AI Engineering with LLMs Specialization (IBM)
- Large Language Model Operations (LLMOps) Specialization, Duke University
- RAG for Generative AI Applications
- Informed Clinical Decision Making using Deep Learning, University of Glasgow

HONORS & AWARDS

1st Place, COMPPEC 2025 | Departmental Silver Medal